

YOUSSEF FELLAH

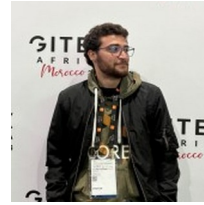
Software Engineer — Backend & Systems Architecture

Distributed Backend (Spring Boot) · Formal Design · InfoSec · Applied Data Science

Seeking Final-Year Internship (PFE) — Scalable Backend Systems & Resilient Architectures

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | y.fellah@mundiapolis.ma



PROFESSIONAL PROFILE

Software Engineering student at Université Mundiapolis specializing in formal analysis, architectural conception (UML 2.5, BPMN 2.0, Clean Architecture, DDD), and distributed systems. Solid engineering foundation centered on the Java/Spring Boot ecosystem for robust backend architectures, augmented by a hybrid profile spanning Information Systems Security (Secure-by-Design, RBAC/IAM, perimeter hardening via OPNsense/VLANs) and Applied Data Science (exploratory analytics, predictive pipelines, and RAG/vector retrieval systems). Rigorous and systems-driven, seeking a Final-Year Internship (PFE) to architect, model, and deploy high-concurrency, secure, and resilient enterprise platforms.

TECHNICAL SKILLS

- **Architecture, Analysis & Conception:** Clean Architecture, Domain-Driven Design (DDD), UML 2.5, BPMN 2.0, Design Patterns (GoF), SOLID Principles, Microservices, REST & RPC APIs
- **Backend Core & Frameworks:** Java (SE/EE), Spring Boot, Spring Security, Spring Data JPA, Hibernate, Node.js, Python, PHP, Solidity, Bash
- **Databases & Data Storage:** PostgreSQL, MySQL, MongoDB, Redis (Caching), Vector DBs, IPFS, SQL Tuning & Schema Normalization
- **Applied Data Science & AI:** Exploratory Data Analysis (EDA), Statistical Validation, Machine Learning Pipelines, Retrieval-Augmented Generation (RAG), Vector Embeddings, LLM Integration, Power BI, Tableau
- **Information Systems Security (InfoSec):** Secure-by-Design Architecture, IAM & RBAC Modeling, Stateless Auth (JWT), Perimeter Hardening (OPNsense, 802.1Q VLANs), OWASP Standards, Endpoint & Protocol Security
- **Cloud & Systems Infrastructure:** AWS (EC2, S3, RDS), Docker, Linux (Fedora/Debian/RHEL), CI/CD (GitHub Actions), Zabbix Telemetry, OpenStack
- **Frontend & Web3 Integration:** Angular, React.js, TypeScript, JavaScript, Tailwind CSS, Web3.js, Ethereum
- **Engineering Governance & Methodologies:** Agile (Scrum / Kanban), Git Flow & Branching Policies, Architecture Decision Records (ADRs), ITIL Fundamentals, ERP Governance (Dolibarr, Odoo)

PROFESSIONAL EXPERIENCE

Software Architect & Systems Engineer (Work-Study)

Feb. 2026 – Aug. 2026

Broker Immobilier — Casablanca, Morocco

- Architected a multi-tenant SaaS real estate CRM utilizing Spring Boot and Angular, applying Clean Architecture principles to enforce domain layer isolation and long-term maintainability.
- Integrated a Retrieval-Augmented Generation (RAG) pipeline and LLM APIs for automated semantic search, document summarization, and metadata extraction.
- Enforced end-to-end security policies using stateless JWT authentication paired with granular Role-Based Access Control (RBAC).

Information Systems & ERP Engineering Intern — Finance

Aug. 2025 – Oct. 2025

HTBS Africa — Casablanca, Morocco

- Analyzed Moroccan tax and treasury compliance regulations, converting operational constraints into formal technical specifications for Dolibarr ERP.
- Developed and deployed 4 custom modular extensions (Treasury, Invoicing, Tax Reporting) adhering to the Open-Closed Principle without modifying the application core.
- Optimized relational database schemas and refactored bottleneck SQL queries, cutting financial report generation time by 40%.

Blockchain Developer Intern — Web3 (Bachelor's Final Project)

Apr. 2025 – Jul. 2025

FST Errachidia — Errachidia, Morocco

- Designed and deployed an Ethereum-based decentralized e-learning and credentials verification platform.
- Authored and audited Solidity smart contracts for immutable academic diploma issuance and integrated IPFS for decentralized file persistence.
- Engineered a responsive Web3 frontend using React.js and Web3.js to handle client-side wallet authentication and contract execution.

MAJOR ENGINEERING PROJECTS

Distributed Architecture: Core Banking — Java, Spring Boot, PostgreSQL

[github.com/ind4x/spring-boot-portfolio](#)

- Designed a decoupled core banking transactional engine isolating ledger operations from edge client APIs; enforced strict ACID transaction boundaries, idempotent processing, and fine-grained RBAC authorization over an integration bus.

Business Modeling & ERP Architecture (Odoo) — BPMN, Python

[github.com/ind4x/si-portfolio](#)

- Architected logistics workflow models for perishable goods via BPMN standards; developed custom Odoo 17 modules adhering to Open-Closed (GoF) principles to eliminate regression risks and ensure full supply-chain traceability.

devctl — Platform Engineering & Local Orchestration — Python, Docker, Bash

[github.com/ind4x/devctl](#)

- Built an asynchronous CLI in Python automating environment bootstrapping, Docker container lifecycle, and dependency orchestration across Spring Boot, Angular, and Vue.js stacks, eliminating cross-environment configuration drift.

Predictive Analytics & Machine Learning Pipelines — Python, Scikit-Learn, Pandas

[github.com/ind4x/machine-learning-portfolio](#)

- Engineered end-to-end classification and clustering pipelines (telecom churn prediction, customer segmentation) applying Random Forest, K-Means, and statistical EDA to generate actionable business intelligence.

Infrastructure Supervision & Firewall Config — AWS, Zabbix, OPNsense

[github.com/ind4x/aws-zabbix-monitoring](#)

- Provisioned multi-OS cloud telemetry on AWS EC2 via Zabbix daemons and Docker; configured an OPNsense firewall enforcing strict 802.1Q VLAN segmentation, protocol security, and zero-trust perimeter traffic filtering.

Decentralized Edu — Solidity, React.js, Web3.js

[github.com/ind4x/decentralized-course-archive](#)

- Authored and deployed gas-efficient Solidity smart contracts on Ethereum for tamper-proof academic credential verification, integrating IPFS via Web3.js and React to persist decentralized, verifiable records.

Software Quality & Engineering Governance — Git, CI/CD, Documentation

- Standardized repository governance by implementing GitHub Actions CI validation policies, strict Trunk-Based development branching rules, and Architecture Decision Records (ADRs) while providing technical mentorship to development teams.

EDUCATION

Engineering Degree (Diplôme d'Ingénieur) in Software Engineering

2025 – Present

Université Mundiapolis — Casablanca, Morocco

- Relevant Focus: Distributed Systems, Advanced Software Architecture, Enterprise Frameworks, Cloud Infrastructure.

Bachelor of Science and Technology (LST) in Software Engineering

2024 – 2025

FST Errachidia — Errachidia, Morocco

- Relevant Focus: Algorithms & Data Structures, Relational Databases, Operating Systems, Web & Mobile Engineering.

CERTIFICATIONS

— AWS Academy Graduate — Cloud Foundations — Amazon Web Services (Nov. 2025) · [Verify](#)

— Advanced Java APIs and Functional Programming — Packt / Coursera (Aug. 2026) · [Verify](#)

— Data Analytics Essentials — Cisco Networking Academy / ENSET (Mar. 2026) · [Verify](#)

— GitHub Foundations (Parts 1 & 2) — Microsoft / GitHub (Apr. 2026) · [Verify](#)

— GitHub Administration Basics and Product Features — Microsoft / GitHub (Apr. 2026) · [Verify](#)

— Introduction to Software Engineering — IBM / Coursera (Feb. 2026) · [Verify](#)

— HCIA-Cloud Computing V5.5 — Huawei ICT Academy

LANGUAGES

Arabic: Native | French: Bilingual / C2 | English: Professional Working Proficiency / C1